Data sources

This project uses **two public datasets** (four CSV files in total). Neither is modified at source; download them yourself and place them in the folders shown below.

1. OpenDengue (version 1.3)

Global database of publicly reported dengue case counts.

- Download page: https://opendengue.org/data.html
- GitHub (all versions): https://github.com/OpenDengue/master-repo
- Data paper: https://doi.org/10.1038/s41597-024-03120-7
- Dataset (Figshare): https://doi.org/10.6084/m9.figshare.24259573
- Licence: CC BY-SA (share adaptations under the same terms)

Files to download (no login needed): on the download page, use the buttons in the "Download global data" section.

File	Button on the website	Used for
`national.csv`	Download global national data	Sections 3–5 (main analysis)
`Spatial_extract_V1_3.csv`	Download global highest spatial resolution data	Section 5.6 (sub-national transfer test)

Do **not** download "highest temporal resolution data"; it is not used.

Check your download (national extract): about 29,873 rows, 129 countries and territories, years 1924–2025. If the numbers differ, a newer version may have been released; use version 1.3 to reproduce the paper.

2. DengAI (DrivenData competition)

Weekly dengue cases for San Juan (Puerto Rico) and Iquitos (Peru). Used only for the external validation in Section 5.9.

- Competition page: https://www.drivendata.org/competitions/44/dengai-predicting-disease-spread/
- Free DrivenData account required. Join the competition, open the **Data** tab and download:

File on the website	Save as
Training Data Features	`dengue_features_train.csv`
Training Data Labels	`dengue_labels_train.csv`

Test Data Features and Submission Format are not used. Please read DrivenData's terms of data use before reusing the data, and cite the competition.

Folder layout

```
project/
├── dengue_revised.ipynb
├── national.csv
├── Spatial_extract_V1_3.csv
└── data/dengai/
    ├── dengue_features_train.csv
    └── dengue_labels_train.csv
```

The notebook reads these paths from environment variables, which you can change:

Variable	Default	Meaning
`DENGUE_CSV`	`national.csv`	OpenDengue national extract
`OUT_DIR`	`results`	Output folder
`DENGAI_DIR`	`data/dengai`	Folder holding the two DengAI files

The spatial extract file name is set by `SPATIAL_RAW_CSV` in the notebook (it is only needed the first time, to build the cached file).

How to cite

- Clarke J, Lim A, Gupte P, Pigott DM, van Panhuis WG, Brady OJ. A global dataset of publicly available dengue case count data. *Scientific Data* 11, 296 (2024). https://doi.org/10.1038/s41597-024-03120-7
- DrivenData. DengAI: Predicting Disease Spread. https://www.drivendata.org/competitions/44/dengai-predicting-disease-spread/